

Chapter 4 – Metasploit Framework

Part 4 – Payloads, Encoders & Listeners

What is a Payload?

A payload is the code that runs after an exploit successfully achieves code execution. Payloads determine what happens next during an authorized assessment.

Payload Types

Single: Self-contained payload.

Staged: Small initial stage followed by a second stage.

Stageless: Entire payload delivered as one unit.

Meterpreter: Advanced interactive payload.

Command Shell: Basic shell access.

Essential Commands

show payloads – List compatible payloads.

set payload PAYLOAD_NAME – Select a payload.

show options – View payload-specific options.

Reverse vs Bind

Reverse payloads initiate a connection back to the tester's listener. Bind payloads wait for the tester to connect. Choose the type appropriate for the authorized lab environment.

Encoders & NOPs

show encoders – List available encoders.

set ENCODER encoder_name – Select an encoder (if supported).

show nops – Display NOP generators.

Encoders modify payload representation but are not a reliable antivirus bypass.

Listeners

use exploit/multi/handler – Load the handler.

set payload ... – Choose the matching payload.

show options – Configure required settings.

run or exploit – Start the handler.

Workflow

Load exploit → Show payloads → Select payload → Configure options → Start handler (if needed) → Execute only in an authorized environment.

Quick Revision

Always select a compatible payload, review payload options, understand reverse vs bind, and ensure the handler matches the payload.